EXP NO: 5 EXPERIMENT TO DETECT OUTLIERS IN A GIVEN DATA SET

AIM:

Understand the procedure to identify the outliers in a given dataset.

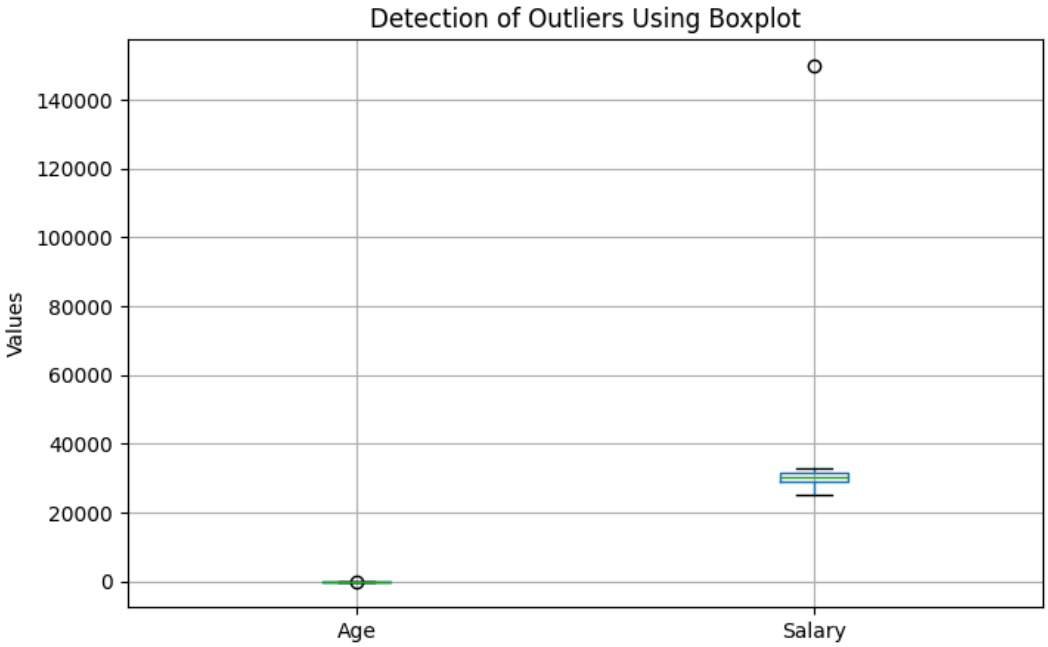
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
 "Age": [20, 21, 22, 23, 21, 24, 22, 25, 23, 22, 21, 100],
 "Salary": [25000, 28000, 30000, 32000, 29000, 31000,
 30500, 33000, 31500, 29500, 28500, 150000]
}
df = pd.DataFrame(data)
for column in df.select_dtypes(include="number").columns:
 Q1 = df[column].quantile(0.25)
 Q3 = df[column].quantile(0.75)
 IQR = Q3 - Q1
 lower_bound = Q1 - 1.5 * IQR
 upper_bound = Q3 + 1.5 * IQR
 outliers = df[
 (df[column] < lower_bound) |
 (df[column] > upper_bound)
 ]
 print("\nColumn:", column)
 print("Q1:", Q1)
 print("Q3:", Q3)
 print("IQR:", IQR)
 print("Lower Bound:", lower_bound)
 print("Upper Bound:", upper_bound)
 print("Outliers:")
 print(outliers[[column]])
plt.figure(figsize=(8, 5))
df.boxplot(column=["Age", "Salary"])
plt.title("Detection of Outliers Using Boxplot")
plt.ylabel("Values")
plt.show()
```

OUTPUT:

```
Column: Age
Q1: 21.0
Q3: 23.25
IQR: 2.25
Lower Bound: 17.625
Upper Bound: 26.625
Outliers:
    Age
11  100

Column: Salary
Q1: 28875.0
Q3: 31625.0
IQR: 2750.0
Lower Bound: 24750.0
Upper Bound: 35750.0
Outliers:
    Salary
11  150000
```



RESULT:

Thus the code is executed successfully.